Verification of the Prophet Secretary Certificate

Abstract

This supplement gives the coefficient construction and the numerical enclosure arguments for the finite certificate accompanying *Mean-only Prophet Inequalities*. It uses the manuscript's original Γ_{p^*}, Lemmas 4.7 and 4.8, and the discretization in Lemmas B.1 and B.2. The supplied rational threshold and proof parameters certify a uniform lower bound of $49839735470020217/72057594037927936 > 0.6916$. We specify the bounds on complete time intervals, the treatment of actual probability averages, and the directed arithmetic used by the verifier.

1 Statement and relation to the manuscript

References to numbered lemmas in the manuscript below refer to the version in which the Prophet Secretary analysis is Section 4 and its finite LP is Appendix B.2. We use the same notation $b, \theta, p^*, P^*, H_{p^*}, \widehat{S}$ and Γ_{p^*}. In particular,

$$\theta(t) = \int_t^1 b(s)\,ds, \qquad P^*(t) = \int_0^t p^*(v)\,dv, \qquad \widehat{S}(s) = (1 - p^*(s)) \exp\!\left(\int_0^s \log(1 - p^*(v))\,dv\right).$$

Here $p^* : [0,1] \to [0,1]$ is any nondecreasing function; it need not be constant on a grid interval or continuous. Integrals are Lebesgue integrals. Values at finitely many partition endpoints do not affect them. As in the manuscript, logarithms at zero and the associated expressions are interpreted by limits.

Theorem 1. *The parameters in* `parameters.json`*, together with the fixed coefficient choices in* `proposals/`*, define an admissible instance of the manuscript's Prophet Secretary construction with*

$$N = 160, \qquad N_t = 5120, \qquad N_p = 10240.$$

For these common parameters, every nondecreasing $p^* : [0,1] \to [0,1]$ and every $\tau \in [0,1]$ satisfy

$$\Gamma_{p^*}(\tau) \;\geq\; \frac{49839735470020217}{72057594037927936} = 0.69166527325054429076\ldots \;>\; \frac{1729}{2500} = 0.6916. \tag{1}$$

The threshold is continuous, concave and strictly decreasing, with

$$\theta(0) = \frac{567246577807}{800000000000} = 0.70905822225875.$$

Consequently, the manuscript's Lemma 4.6 yields the claimed 0.6916 *competitive ratio in Theorem 1.2 for every n.*

The finite arithmetic checks used in this theorem are specified below and can be repeated with the included verifier. The certificate establishes feasibility, not optimality of the threshold or of the finite LP. It concerns the Prophet Secretary lower bound; it is not a certificate for the manuscript's separate i.i.d. or upper-bound results.

2 Common data, grids and fixed plane choices

The manuscript indexes drop rates by $1, \dots, N$. For zero-based file index $a \in \{0, \dots, N-1\}$, put $J_a = [a/N, (a+1)/N)$. The functions b, m, κ are constant on J_a, and

$$b(s) = b_{a+1} = \frac{\texttt{b_numerators}[a]}{10^{10}},$$
$$m(s) = \frac{\texttt{m_numerators}[a]}{10^{10}}, \qquad \kappa(s) = \frac{\texttt{kappa_numerators}[a]}{10^{10}}.$$

The auxiliary function m specifies the allocation between the two original extra-gain bounds; its precise use is given in Section 3. The verifier checks

$$0 < b_1 < \cdots < b_N, \qquad \frac{1}{N}\sum_{a=1}^{N} b_a \le 1, \qquad 0 \le m(s) \le b(s), \qquad \kappa(s) \ge 0.$$

Every proposal contains the same 640 common integer parameters. The first three blocks specify these three functions; the fourth block is checked to be identically zero and contributes no term.

Write

$$h = \frac{1}{N_p}, \qquad s_j = jh, \qquad I_j = [s_j, s_{j+1}], \qquad \xi_j = \frac{1}{h}\int_{I_j} p^*(s)\,ds \quad (0 \le j < N_p).$$

Monotonicity gives $0 \le \xi_0 \le \cdots \le \xi_{N_p-1} \le 1$. The time intervals are

$$T_\ell = [t_\ell^-, t_\ell^+] = [2\ell h, (2\ell+2)h], \qquad 0 \le \ell < N_t.$$

They cover $[0,1]$. Each lies within a single constant-slope piece of θ. The probability partition refines all breakpoints of b, m, κ and the route functions used below.

Each proposal has an additional plane-owner partition into equal cells. Its size is 160, 320 or 640 in this release, and divides N_p. It specifies three types of plane lists: ordinary H planes, capped H planes, and $\widehat{S}$ planes. A plane has a nonnegative integer mass with denominator 2^{32}. The total mass is checked to be at most one separately for every owner and every type. Those masses and the plane slopes or ρ values are fixed on the entire T_ℓ. The two stored H slope integers represent σ and ν with denominator 2^{32}; a stored survival-plane parameter represents ρ with denominator 10^{10}.

If $f \ge 0$ and $f \ge L_i$ pointwise, then, for $\omega_i \ge 0$ with $\sum_i \omega_i \le 1$,

$$f \ge \sum_i \omega_i L_i.$$

Indeed, give unused mass to the zero lower bound. This applies to both H_{p^*} and $\widehat{S}$, which are nonnegative. For the former, when its logarithms are finite,

$$\mathcal{L}(d, x, y) = -\log\left(1 - \frac{dx}{1-y}\right) - dx \ge 0$$

by $-\log(1-z) \ge z$ for $0 \le z < 1$; the limiting cases follow as well. Thus the plane mixtures are fixed affine lower bounds of exactly the kind permitted by Lemma B.1. They do not mix algorithms or thresholds.

3 Allocation and comparison contributions

3.1 Fixed allocation and admissibility

The endpoint and starting-point functions are fixed independently of p^*. For $s \in J_a$, define

$$u(s) = r_a + (1 - r_a)s, \qquad z(s) = q_a s,$$

where the following table specifies every route used by the verifier. The first column is $\lfloor a/4 \rfloor$, rather than the threshold-segment index a itself.

$\lfloor a/4 \rfloor$	r_a	q_a
16	3/4	1
17	7/10	9/10
$18, \dots, 27$	1/2	1/2
32	1/4	1/2
$33, \dots, 37$	1/4	1/4
all remaining indices	0	1

In particular, $s \le u(s) \le 1$ and $0 \le z(s) \le s$.

In the notation of the article, use two applications of Lemma 4.3 and one comparison application of Lemma 4.4:

$$\begin{aligned} c_1(s) &= 1, & \lambda_1(s) &= b(s) - m(s), \\ c_2(s) &= u(s), & \lambda_2(s) &= m(s), \\ z_1(s,t) &= z(s), & \mu_1(s,t) &= m(s)\,\mathbf{1}_{\{t > u(s)\}}. \end{aligned}$$

The strict inequality in the last indicator ensures that the allocation constraint holds also at $t = u(s)$, where the capped application of Lemma 4.3 still uses its coefficient. For every $s \le t < 1$, the total allocated coefficient is exactly $b(s)$: it is $(b(s) - m(s)) + m(s)$ before and at $u(s)$, and $(b(s) - m(s)) + m(s)$ after $u(s)$, with the second summand coming from the comparison application in the latter case. Hence (4.1) holds pointwise. Changing the value of an indicator at $t = u(s)$ leaves all integrals below unchanged.

For these data, $m(s)$ is positive precisely on the union of the threshold segments with indices $64 \le a \le 111$ and $136 \le a \le 151$. The route table consequently gives

$$m(s) > 0 \quad \Longrightarrow \quad z(s) \ge \frac{17}{80}.$$

In particular, the comparison denominator satisfies $1 - t + z(s) \ge 17/80$ wherever the comparison weight is nonzero. Since m is bounded, the comparison integrability condition in Appendix B.2 holds. All changes in the order of integration below are therefore justified; for the nonnegative terms they also follow directly from Tonelli's theorem. Expressions multiplied by $m(s) = 0$ are taken to contribute zero, so no value at an irrelevant zero denominator is needed.

3.2 Exact comparison kernels and conservative cell coefficients

Write $P^*(v) = \int_0^v p^*(w)\,dw$. With the allocation just specified, the comparison contribution to $\Gamma_{p^*}(\tau)$ is

$$\int_0^1 m(s) \int_{u(s)}^1 \left(\mathbf{1}_{\{\tau \le t\}} p^*(s) + \frac{P^*(t) - P^*(z(s)) - \mathbf{1}_{\{\tau \le t\}}}{1 - t + z(s)} \right) dt\,ds. \tag{2}$$

The equality uses only the preceding allocation and the article's definition of Γ_{p^*}. Together with

$$F_{p^*}(\tau) = \tau + \theta(\tau) - \int_0^1 b(s) \min\{s, \tau\} p^*(s)\,ds,$$

the complete instantiated criterion consists of $F_{p^*}(\tau)$, the comparison contribution (2), and

$$\begin{aligned}&\int_0^1 (b(s)-m(s))H_{p^*}(s,(\tau-s)_+)\,ds\\&\quad+\int_0^1 m(s)H_{p^*}(s,(\min\{\tau,u(s)\}-s)_+)\,ds\\&\quad+\int_0^1 \kappa(s)\widehat{S}(s)\,ds-\int_0^\tau \kappa(s)(1-p^*(s))\,ds.\end{aligned}$$

Thus every term of the original criterion is accounted for below.

For $0\le v\le 1$, define the nonnegative kernel

$$K(v)=\int_0^1 m(s)\log\Big(\frac{1-\max\{v,u(s)\}+z(s)}{z(s)}\Big)\,ds. \tag{3}$$

Indeed, the logarithm equals $\int_{\max\{v,u(s)\}}^1 (1-t+z(s))^{-1}\,dt$. It is nonincreasing in v for each fixed source point s.

Fix one stopping-time interval $T_\ell=[t_\ell^-,t_\ell^+]$ and a probability cell $I_j=[s_j,s_{j+1}]$, of length $h=1/N_p$ with $N_p=10240$. The common probability partition refines the threshold and route partitions. On I_j, use the affine extension of its own route and put

$$\overline{u}_j=r_a+(1-r_a)s_{j+1}\quad\text{when the interior of } I_j \text{ lies in } J_a.$$

This definition uses the route on the interior of I_j, even if the next cell uses a different route. Thus $u(s)\le\overline{u}_j$ for almost every $s\in I_j$. As in the article, write

$$\xi_j=\frac{1}{h}\int_{I_j} p^*(s)\,ds.$$

The first term in (2) integrates exactly to $\int_0^1 m(s)(1-\max\{\tau,u(s)\})p^*(s)\,ds$. For every $\tau\in T_\ell$, its contribution from I_j is at least

$$h\,m(s_j)(1-\max\{t_\ell^+,\overline{u}_j\})\xi_j. \tag{4}$$

Here and below the step functions are represented by their values on the cell interior, with the right-continuous convention at s_j. The term with the negative indicator is exactly $-K(\tau)$. Since K is nonincreasing, it is at least $-K(t_\ell^-)$ throughout T_ℓ.

For the remaining, nonnegative comparison-history term, use $P^*(t)-P^*(z(s))=\int_{z(s)}^t p^*(v)\,dv$ and exchange the integrals. The result is

$$\int_0^1 p^*(v)\left[\int_{\{s:z(s)\le v\}} m(s)\log\Big(\frac{1-\max\{v,u(s)\}+z(s)}{z(s)}\Big)\,ds\right]dv. \tag{5}$$

For $v\in I_j$, retain only the source points with $z(s)\le s_j$. They are eligible for every v in that cell. On this fixed source set, replace v by s_{j+1} inside the nonincreasing logarithmic kernel. Define

$$K_j=\int_{\{s:z(s)\le s_j\}} m(s)\log\Big(\frac{1-\max\{s_{j+1},u(s)\}+z(s)}{z(s)}\Big)\,ds. \tag{6}$$

Then (5) is at least $h\sum_j K_j\xi_j$. This argument does not assume that the expression in square brackets in (5) is nonincreasing: its source set grows with v. The source set in (6) is fixed before taking a cell endpoint.

Combining the three contributions, a lower bound on (2), uniform over $\tau\in T_\ell$, is

$$-K(t_\ell^-)+h\sum_{j=0}^{N_p-1}\Big[m(s_j)(1-\max\{t_\ell^+,\overline{u}_j\})+K_j\Big]\xi_j. \tag{7}$$

Every coefficient is fixed on the whole stopping-time interval and is independent of p^*.

Evaluation of the logarithmic kernels. On a source segment with route constants $r = r_a$ and $q = q_a$, put $d = (v - r)/(1 - r)$. In the portion with $s \le d$ the integrand in (3), without its factor $m(s)$, is

$$\log \frac{1 - v + qs}{qs};$$

in the portion with $s \ge d$ it is

$$\log \frac{1 - r + (q + r - 1)s}{qs}.$$

The splitting point is clipped to the source integration interval. For (6), that interval is additionally cut off at $s = s_j/q$. If $v = 1$, both kernel integrals vanish directly. For $v < 1$, an antiderivative of $\log((A + Cs)/(qs))$ is

$$\begin{cases} \dfrac{(A + Cs)(\log(A + Cs) - 1)}{C} - s(\log(qs) - 1), & C \neq 0, \\ s \log A - s(\log(qs) - 1), & C = 0. \end{cases} \tag{8}$$

Differentiating proves the formula wherever the logarithmic arguments are positive. The two branches use $(A, C) = (1 - v, q)$ and $(A, C) = (1 - r, q + r - 1)$, respectively. Endpoint limits use $x \log x \to 0$ as $x \downarrow 0$; in particular the antiderivative at $s = 0$ is $A(\log A - 1)/C$ when $A > 0$ and $C \neq 0$, and is zero in the remaining applicable limiting cases. These are exactly the branches evaluated by `primitive` and `integrate`.

4 The logarithmic contributions on complete rectangles

Let $p^* : [0, 1] \to [0, 1]$ be nondecreasing, and use the paper's definitions of P^*, H_{p^*}, and $\mathcal{L}$. In particular,

$$P^*(s) = \int_0^s p^*(v)\, dv, \qquad \mathcal{L}(\Delta, x, y) = \log(1 - y) - \log(1 - \Delta x - y) - \Delta x.$$

For $0 \le \Delta \le 1 - s$, the manuscript defines

$$H_{p^*}(s, \Delta) = \mathcal{L}\left(\Delta, p^*(s), \int_{s-\delta}^{s} p^*(v)\, dv\right), \qquad \delta = \min\{s, (1 - \Delta)/2\}.$$

The case $\Delta = 0$ is interpreted as $H_{p^*}(s, 0) = 0$. Fix a probability cell $I_j = [s_j, s_{j+1}]$, where $s_i = ih$ and $h = 1/N_p$, and a complete stopping-time interval $T_\ell = [t_\ell^-, t_\ell^+]$. All bounds in this section hold for every $s \in I_j$ and every $\tau \in T_\ell$. At a route breakpoint, endpoint values of the route mean the extensions of its formula on the cell under consideration; the choices at individual breakpoints do not affect any integral.

4.1 Duration and retained history

For the ordinary logarithmic contribution, the duration is $(\tau - s)_+$. For the capped contribution, it is $(\min\{\tau, u(s)\} - s)_+$, where $s \le u(s) \le 1$ and the fixed route u is nondecreasing on the cell. For the contribution being considered, denote its duration by $\Delta(s, \tau)$ and set

	d	d_{up}
ordinary	$(t_\ell^- - s_{j+1})_+$	$(t_\ell^+ - s_j)_+$
capped	$(\min\{t_\ell^-, u(s_j)\} - s_{j+1})_+$	$(\min\{t_\ell^+, u(s_{j+1})\} - s_j)_+$

Monotonicity of the positive-part and minimum operations gives

$$0 \le d \le \Delta(s, \tau) \le d_{\mathrm{up}} \le 1.$$

If $d = 0$, the verifier uses the zero lower bound for this contribution. This is valid because, wherever the logarithms are finite,

$$\mathcal{L}(\Delta, x, y) = -\log\left(1 - \frac{\Delta x}{1-y}\right) - \Delta x \geq \frac{\Delta x}{1-y} - \Delta x \geq 0,$$

and the same conclusion holds at the extended-real boundary. We therefore consider $d > 0$ below.

In the definition of $H_{p^*}(s, \Delta(s,\tau))$, the history length is

$$\delta(s,\tau) = \min\left\{s, \frac{1-\Delta(s,\tau)}{2}\right\}.$$

Define the common lower bound

$$\delta_0 = \min\left\{s_j, \frac{1-d_{\mathrm{up}}}{2}\right\}, \qquad k = \min\left\{j, \max\left\{0, \lceil N_p(s_{j+1} - \delta_0)\rceil\right\}\right\}.$$

The verifier retains the cells with indices $k, \ldots, j-1$, and puts

$$w = (j-k)h, \qquad y = \int_{s_k}^{s_j} p^*(v)\,dv = h\sum_{i=k}^{j-1}\xi_i, \qquad \xi_i = \frac{1}{h}\int_{I_i} p^*(s)\,ds.$$

When $k = j$, the retained history is empty and both w and y are zero; the displayed integral and sum are understood to be zero. In particular, clipping a proposed starting index larger than j does not retain an earlier nonempty interval.

If $k < j$, the unclipped ceiling is at most $j-1$, and

$$s_k \geq s_{j+1} - \delta_0, \qquad w = s_j - s_k \leq \delta_0.$$

Furthermore, $\delta(s,\tau) \geq \delta_0$. Hence

$$s - \delta(s,\tau) \leq s_{j+1} - \delta_0 \leq s_k \leq s_j \leq s.$$

Thus the retained history lies inside the history in the definition of H_{p^*}, simultaneously throughout the rectangle. In the empty case the same integral inequality holds directly. We obtain

$$0 \leq y \leq \int_{s-\delta(s,\tau)}^{s} p^*(v)\,dv, \qquad 0 \leq y \leq w\,p^*(s).$$

The second inequality uses the monotonicity of the original function p^*, since every retained point precedes s.

The logarithmic expression is nondecreasing in its duration and history arguments. Indeed, on its finite domain,

$$\frac{\partial \mathcal{L}}{\partial \Delta} = \frac{x}{1-\Delta x - y} - x \geq 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = \frac{1}{1-\Delta x - y} - \frac{1}{1-y} \geq 0.$$

Consequently,

$$H_{p^*}(s, \Delta(s,\tau)) \geq \mathcal{L}(d, p^*(s), y).$$

This inequality also holds when an original logarithm is singular, by its limiting interpretation; such a value is $+\infty$ and causes no difficulty for a finite lower bound.

The reduced triangle satisfies the strict domain requirement of Lemma 4.7. If $d_{\mathrm{up}} < 1$, then

$$d + w \leq d_{\mathrm{up}} + \frac{1-d_{\mathrm{up}}}{2} = \frac{1+d_{\mathrm{up}}}{2} < 1.$$

If $d_{\mathrm{up}} = 1$, then $\delta_0 = w = 0$, while $d \leq (t_\ell^- - s_{j+1})_+ < 1$. Thus $d + w < 1$ also in this case. Lemma 4.7 therefore applies with $\Delta = d$, $\delta = w$, and

$$D = \{(x,y) : 0 \leq x \leq 1,\ 0 \leq y \leq wx\}.$$

4.2 Recomputing the intercept by the paper's edge tangents

Fix the proposed slopes $\sigma, \nu \in \mathbb{R}$. The verifier recomputes an intercept for the reduced triangle, rather than reusing an intercept from another rectangle. For completeness, write

$$R(x, y) = \mathcal{L}(d, x, y) - \sigma x - \nu y.$$

If $w > 0$, then R is continuous on D and

$$\det \nabla^2 R(x, y) = -\frac{d^2}{(1-y)^2(1-dx-y)^2} < 0$$

at each interior point. An interior local minimum would have a positive semidefinite Hessian, so a global minimum occurs on the boundary. Parametrize the three edges by

$$(x_0(v), y_0(v)) = (v, 0), \qquad (x_1(v), y_1(v)) = (v, wv), \qquad (x_2(v), y_2(v)) = (1, wv), \quad 0 \le v \le 1,$$

and put $\phi_e(v) = R(x_e(v), y_e(v))$. As in the paper,

$$\begin{aligned}
\phi_0''(v) &= \frac{d^2}{(1-dv)^2} \ge 0, \\
\phi_1''(v) &= \frac{(d+w)^2}{(1-(d+w)v)^2} - \frac{w^2}{(1-wv)^2} \ge 0, \\
\phi_2''(v) &= \frac{w^2}{(1-d-wv)^2} - \frac{w^2}{(1-wv)^2} \ge 0.
\end{aligned}$$

For each edge, choose any $v_e \in [0, 1]$. Convexity gives the uniform tangent lower bound

$$\phi_e(v) \ge \phi_e(v_e) + \phi_e'(v_e)(v - v_e), \qquad 0 \le v \le 1.$$

It follows that

$$\beta_e = \phi_e(v_e) - v_e \phi_e'(v_e) + \min\{0, \phi_e'(v_e)\}, \qquad \beta = \min_{e \in \{0,1,2\}} \beta_e$$

is a valid intercept. The selected tangent point need not minimize ϕ_e; its only required property is $v_e \in [0, 1]$. If $w = 0$, the triangle is the segment $(v, 0)$ and edge 0 suffices. Including the duplicated or collapsed edges can only replace the intercept by another valid lower bound.

We have proved, for the fixed rectangle and plane,

$$H_{p^*}(s, \Delta(s, \tau)) \ge \beta + \sigma p^*(s) + \nu h \sum_{i=k}^{j-1} \xi_i.$$

Averaging over the probability cell gives

$$\frac{1}{h} \int_{I_j} H_{p^*}(s, \Delta(s, \tau))\, ds \ge \beta + \sigma \xi_j + \nu h \sum_{i=k}^{j-1} \xi_i, \qquad \tau \in T_\ell.$$

All averages here are averages of the actual p^*. No step-function replacement of p^* has been made. Multiplication by the fixed nonnegative allocation on the cell and by any fixed nonnegative plane mass preserves this inequality.

5 The survival contribution and its coefficient error

Fix $I_j = [s_j, s_{j+1}]$ and a plane parameter $\rho \in [0, 1]$. Apply Lemma 4.8 with $c = s_{j+1} > 0$, and abbreviate

$$A = \rho^{s_{j+1}}(1 + s_{j+1} - s_{j+1}\rho), \qquad B = \rho^{1+s_{j+1}}.$$

With the paper's notation $\widehat{S}(s)$, the resulting bound is

$$\widehat{S}(s) \geq A(1 - p^*(s)) - BP^*(s), \qquad s \in I_j.$$

Both constants are nonnegative. When $\rho = 0$, they are zero and the following formulas remain valid.

Fubini's theorem on the bounded cell gives the exact identity

$$\frac{1}{h}\int_{s_j}^{s_{j+1}} P^*(s)\, ds = h\sum_{i<j}\xi_i + \frac{1}{h}\int_{s_j}^{s_{j+1}} (s_{j+1} - v)p^*(v)\, dv.$$

For the last term, use the coefficient-error inequality in the proof of Lemma B.1 with coefficient

$$k(v) = -\frac{B}{h}(s_{j+1} - v), \qquad d_j = -\frac{B}{2}.$$

Specifically, $0 \leq p^* \leq 1$ implies

$$\int_{I_j} k(v)p^*(v)\, dv \geq hd_j\xi_j - \int_{I_j}(d_j - k(v))_+\, dv.$$

The error is exactly

$$\begin{aligned}\int_{I_j}(d_j - k(v))_+\, dv &= \frac{B}{h}\int_{s_j}^{s_j+h/2}(s_j + h/2 - v)\, dv \\ &= \frac{Bh}{8}.\end{aligned}$$

Combining these expressions proves the cell-average lower bound

$$\boxed{\frac{1}{h}\int_{I_j}\widehat{S}(s)\, ds \geq A - \frac{Bh}{8} - \left(A + \frac{Bh}{2}\right)\xi_j - Bh\sum_{i<j}\xi_i.}$$

Thus a fixed nonnegative correction allocation and plane mass add their product with $h(A - Bh/8)$ to the constant, subtract their product with $h(A + Bh/2)$ from the current-cell coefficient, and subtract their product with Bh^2 from every preceding-cell coefficient. This uses exactly the coefficient-error inequality stated in Lemma B.1, including its $Bh/8$ deduction before multiplication by the cell length and allocation.

6 Collecting coefficients and verifying the original LP

For a fixed time interval T_ℓ, the preceding bounds give exact real coefficients $C_\ell, g_{\ell,0}, \ldots, g_{\ell,N_p-1}$ such that

$$\Gamma_{p^*}(\tau) \geq C_\ell + \sum_{j=0}^{N_p-1} g_{\ell,j}\xi_j \qquad (\tau \in T_\ell). \tag{9}$$

All plane parameters and route functions are fixed. These coefficients are linear in the common drop rates and proof-weight coefficients, with the baseline variable v_ℓ as in Lemma B.1. The common rational input is one candidate solution of that original finite LP.

Here are the remaining baseline and prefix bounds used in collecting (9). Since $\tau + \theta(\tau)$ is affine on T_ℓ, its minimum is

$$v_\ell = \min\{t_\ell^- + \theta(t_\ell^-), t_\ell^+ + \theta(t_\ell^+)\}.$$

For the negative part of F_{p^*}, b is constant on I_j, so

$$-\int_{I_j} b(s)\min\{s,\tau\}p^*(s)\,ds \geq -hb(s_j)\min\{s_{j+1}, t_\ell^+\}\xi_j.$$

For the correction prefix, use $\kappa(1-p^*) \geq 0$ to obtain

$$\begin{aligned}-\int_0^\tau \kappa(s)(1-p^*(s))\,ds &\geq -\int_0^{t_\ell^+} \kappa(s)(1-p^*(s))\,ds \\ &= -\int_0^{t_\ell^+} \kappa(s)\,ds + \sum_{j:\ s_{j+1}\leq t_\ell^+} h\kappa(s_j)\xi_j.\end{aligned}$$

The last equality uses the aligned grid, so no within-cell approximation is present.

For completeness, Lemma B.2 eliminates the ordered averages exactly. Set $\xi_{-1} = 0$ and $\xi_{N_p} = 1$. Then

$$C_\ell + \sum_j g_{\ell,j}\xi_j = \sum_{k=0}^{N_p}(\xi_k - \xi_{k-1})\left(C_\ell + \sum_{j=k}^{N_p-1} g_{\ell,j}\right).$$

The coefficients form a convex combination. Conversely each suffix value is attained by setting $\xi_j = \mathbf{1}_{\{j\geq k\}}$. Hence the minimum is

$$C_\ell + \min_{0\leq k\leq N_p} \sum_{j=k}^{N_p-1} g_{\ell,j}. \tag{10}$$

The empty suffix $k = N_p$ is included. The verifier evaluates every suffix in every time interval, a total of $5120 \cdot 10241 = 52\,433\,920$ suffix constraints.

The directed arithmetic described next supplies lower endpoints for C_ℓ and every $g_{\ell,j}$. Since $\xi_j \geq 0$, substituting those lower endpoints into (9) remains a lower bound. Thus the numerical check certifies all exact original LP constraints at the supplied rational candidate; it does not require treating floating-point solver output or rounded coefficients as a different mathematical LP.

7 Directed arithmetic and certification of the numerical bounds

All input parameters used in the proof are rational. Set

$$Q = 2^{56}, \qquad G = 2^{62}.$$

The verifier represents an interval by two integers (a, b), with $-G < a \leq b < G$, denoting $[a/Q, b/Q]$. A rational number is enclosed by rounding its scaled value down and up. Addition and subtraction use the corresponding endpoint formulas. Multiplication takes the minimum and maximum of the four endpoint products and divides by Q, rounding outwards. Division similarly considers the four endpoint quotients; it is used only when the denominator interval has strictly positive lower endpoint. For a signed numerator and a positive denominator, the integer routines compute the mathematical floor and ceiling, including when the numerator is negative. Consequently, each operation encloses the result for every choice of real operands from its input intervals. Shared occurrences of the same quantity can enlarge the resulting interval, but cannot invalidate this containment statement.

Logarithms. For a positive dyadic endpoint x, exact powers of two reduce x to $x = 2^k r$, where $1 \le r \le 2$. The verifier evaluates $\log x = k \log 2 + \log r$ with directed arithmetic. Writing $z = (r-1)/(r+1)$ gives $0 \le z \le 1/3$ and

$$\log r = 2 \sum_{q=0}^{23} \frac{z^{2q+1}}{2q+1} + R, \qquad 0 \le R \le \frac{2z^{49}}{49(1-z^2)} \le \frac{9}{196\,3^{49}} < 2^{-79} < Q^{-1}.$$

Indeed, the expansion follows by integrating the geometric series for $2/(1-z^2)$ from 0 to z; bounding every remaining denominator by 49 gives the displayed remainder bound. The final strict numerical inequality is the integer inequality $9\,2^{79} < 196\,3^{49}$. The code evaluates the first 24 terms by interval operations and adds one integer unit to the upper endpoint. This includes the omitted nonnegative remainder. The same procedure at $r = 2$ encloses $\log 2$. For a positive interval, monotonicity of the logarithm allows its lower and upper endpoints to be evaluated separately. Boxing a positive rational before this step therefore also gives a valid enclosure of its logarithm.

Exponentials and powers. To evaluate e^x at an endpoint $x \le 0$, choose $k \ge 0$ so that $z = -x/2^k \in [0, 1/2]$. The positive Taylor polynomial and its remainder obey

$$e^z = \sum_{n=0}^{20} \frac{z^n}{n!} + R, \qquad 0 \le R \le \frac{e^{1/2}(1/2)^{21}}{21!} < 2^{-84} < Q^{-1}.$$

Here $e^{1/2} < 2$, for example by termwise comparison of its exponential series with the geometric series, and $21! > 2^{64}$; these two inequalities give the strict 2^{-84} bound. The verifier adds one integer unit to the upper endpoint of the polynomial enclosure, takes its reciprocal, and then squares k times. Each step is an enclosing interval operation, so the result contains e^x. Monotonicity supplies the enclosure for an interval of nonpositive exponents. For $0 < \rho \le 1$ and $0 \le c \le 1$, the expression ρ^c is evaluated as $\exp(c \log \rho)$. The exact cases $\rho = 1$, $c = 0$, and $\rho = 0 < c$ are handled directly. The survival coefficient ρ^{1+c} is computed as $\rho\,\rho^c$.

Choice of tangent points and weighted planes. In the proof of Lemma 4.7, let ϕ_e denote the restriction of $\mathcal{L}(\Delta, x, y) - \sigma x - \nu y$ to one of the three edges of the triangle. For every chosen rational $u_e \in [0, 1]$, convexity gives

$$\min_{0 \le u \le 1} \phi_e(u) \ge \phi_e(u_e) - u_e \phi_e'(u_e) + \min\{0, \phi_e'(u_e)\}.$$

The verifier encloses the right-hand side and uses its lower endpoint. Taking the minimum of the three resulting lower endpoints gives a certified intercept β. A floating-point derivative search is used only to suggest the rational point u_e (with denominator 2^{30}). The displayed inequality holds for every point in $[0, 1]$; neither root-finding accuracy nor a floating-point assertion about an optimum is a premise of the proof.

If a rational slope is too large to be boxed at the chosen scale, the code instead certifies the already weighted expression. For the exact rational weight $\omega \ge 0$, it encloses $\omega\sigma$ and $\omega\nu$ directly and applies the same tangent formula to $\omega\phi_e$. Multiplication by a nonnegative constant preserves convexity, and the three-edge argument in Lemma 4.7 remains valid. Thus the resulting plane has the form

$$\omega \mathcal{L}(\Delta, x, y) \ge \beta_{\mathrm{w}} + \omega\sigma x + \omega\nu y.$$

Here β_{w} is the certified rational weighted intercept. When $\omega > 0$, dividing by ω gives the fixed valid unweighted intercept $\beta = \beta_{\mathrm{w}}/\omega$; a zero-weight contribution is discarded. Thus this computation also specifies a fixed plane that can be multiplied by a variable proof weight in the manuscript's linear LP. This is a scaled application of Lemma 4.7, rather than an additional nonlinear lower-bound family. Enclosing the exact weight at each arithmetic operation, even when occurrences are treated independently, preserves containment of the exact weighted formula.

Integer ranges for the supplied certificate. The range claim concerns the finite input files in this release. It does not assert that arbitrary malformed or adversarial input is accepted safely. Their grids have $N_p = 10240$ and owner counts in $\{160, 320, 640\}$, and each proposal has at most 1972 planes. All interval endpoints are checked against G. Products of two such endpoints have magnitude less than 2^{124}, and an endpoint multiplied by Q has magnitude less than 2^{118}; both fit in a signed 128-bit integer. Rational-to-interval scaling additionally has an explicit overflow check. The rational routines cancel common factors and reject results outside their 64-bit range. Products of two 64-bit numerators or denominators, and the two-product sums used in rational addition, fit in signed 128-bit arithmetic for these inputs.

The few integer accumulators outside the interval class also remain in range. For example, there are at most $1972 \cdot 64$ range updates per rectangle, with each update endpoint bounded in magnitude by G; even counting both endpoint events, summing their absolute values, and then multiplying by N_p gives a bound less than 2^{94}. Suffix sums are bounded by $N_p G < 2^{76}$. The exact common parameter numerators are at most 20808318421 with denominator 10^{10}, so the baseline and correction-prefix accumulations also fit easily in 128 bits. In logarithm range reduction, the dyadic endpoints are in $(0, G)$, and the reduced numerator plus denominator is less than $96Q < 2^{63}$. The exponential range reduction requires a divisor of at most 128 under the same guard. The completed replay checks every rectangle without a range exception. These finite range facts, together with the directed formulas above, justify the arithmetic used for this certificate.

Final comparison. The assembled coefficient intervals enclose a valid affine lower bound from Lemma B.1. Replacing each coefficient by its lower endpoint preserves a lower bound because every interval average lies in $[0, 1]$. Lemma B.2 then reduces its minimum to the minimum of all suffix sums, including the empty suffix. If the resulting lower integer is v, the target test is the exact integer comparison

$$2500v \geq 1729Q.$$

No rounded decimal, numerical optimization status, or floating-point comparison is used to accept the competitive-ratio guarantee.

8 Reproduction and the finite verification result

From the directory containing `replay.py`, run

```
python3 replay.py --audit-only
python3 replay.py --jobs 8
```

Use Python 3.9 or later and a GNU-compatible C++17 compiler supporting ___int128. No external Python package is needed. The first command checks file hashes, common parameters, plane-mass budgets, and complete coverage. The second compiles the verifier and computes all 5120 arithmetic records. It checks the exact target 1729/2500 for each interval and the exact minimum below. Records are generated in the output directory.

The minimum reconstructed numerator at denominator 2^{56} is 49839735470020217. It occurs on

$$T_\ell = [10222/10240, 10224/10240],$$

with minimizing suffix index 20. Its exact margin above the target is

$$\frac{49839735470020217}{72057594037927936} - \frac{1729}{2500} = \frac{2939645868285289}{45035996273704960000} > 0.$$

The complete replay passes, and every time interval has lower bound at least this minimum. Combining (9)–(10) with this finite computation proves (1). Positivity and monotonicity of the

supplied drop rates give the asserted properties of θ. Finally, Lemma 4.6 of the manuscript transfers the uniform bound on $\Gamma_{p^*}(\tau)$ to the expected reward of the algorithm.

Files implementing the proof.

sources/strict160_verify.cpp constructs the coefficients on complete rectangles and enumerates all suffixes. sources/log_plane.hpp certifies the three-edge tangent intercepts; sources/interval.hpp supplies directed arithmetic. The exact common parameters are in parameters.json, and proposals/ contains the 5120 fixed plane lists. replay.py assigns these lists to the consecutive time intervals defined in Section 2 and computes fresh verification records. certificate.json contains the integrity hashes.

The auxiliary tail field in each proposal is parsed but does not determine the route. The route is exactly the one specified in Section 3 and in the evaluator. Verification uses the fixed rational proposals directly and does not require the search that discovered them.